MAVERICK

USER GUIDE

AI-Driven Anomaly Detection in Component Burn-In & Screening

Department of Space | ISRO | Smart India Hackathon 2026

Version 1.0 | For QA Inspectors, Electronic Test Engineers & Evaluators
This guide is for first-time users who have never used the web application.

1. About the System

MAVERICK is a web application that helps find defective electronic components (ICs / chips / circuit parts) that pass the usual "static limit" screening but still show hidden abnormal behaviour during burn-in testing. Burn-in means running the component at a high temperature (for example 125°C) for a long time, and measuring its electrical parameters at different hours — **0h, 24h, 96h and 168h**.

The system has two AI tools (called Modules):

- **Module A - Dynamic Outlier Detection** : Compares each component against its own manufacturing *lot*. Even if a part is inside the datasheet limit (for example below 50µA), it is flagged if it drifts far away from the average of its lot (e.g. lot average is 10µA but this part shows 45µA).
- **Module B - Time-Series Drift Predictor** : Takes the 0h and 24h readings, forecasts the 168h value, and checks whether the predicted drift is faster than a safe limit. If it is, the component is flagged for early rejection.

Why this matters

A defective part that escapes screening can fail catastrophically inside a satellite or rocket payload. This system reduces those escapes by catching "latent defects" early and by explaining every decision to the QA inspector.

2. How to Start the Application

2.1 Starting the server (on the deployment PC)

1. Open the project folder on your computer.
2. Double-click `start.bat`. Two windows open automatically:
   - The **Backend server** window (FastAPI – processes the data).
   - The **Frontend server** window (Next.js – the webpage you see).
3. Wait for the message `Ready` in the frontend window.

2.2 Opening the webpage

1. Open any browser (Chrome / Edge / Firefox).
2. Go to: `http://localhost:3000`
3. If you are testing from another computer on the same network, use the shared link given with the deployment (the Cloudflare tunnel URL).

3. How to Log In

On the first screen you will see the login page with a **secure gateway** design.

1. Enter your **Email** and **Password** in the fields.
2. Click the **Login** button.
3. If a field is empty or the password is wrong, a red message appears. Check and try again.

Demo accounts (use any one of these)

Role	Email	Password	What this account can see
Admin	`admin@isro.gov.in`	`Admin@123!`	Everything
QA Inspector	`qa@isro.gov.in`	`Qa@123!`	Everything
Engineer	`engineer@isro.gov.in`	`Eng@123!`	Everything

Passwords are strong on purpose. New registrations must also use a strong password (at least 8 characters, upper-case letter, lower-case letter, number and a special symbol).

After login you land on the **Dashboard (Mission Control)** page. On the left there is a menu (sidebar) with the pages: **Dashboard, Upload Data, Analysis Center** and your **account info**.

4. Step-by-Step: How to Test the System

Follow these steps in order. The whole test takes about 2–3 minutes.

1 **Load data** — go to **Upload Data** in the left menu.

You have two simple ways:

- **Option 1 (easiest) - Sample Data tab:** click the **Generate & Load Sample Data** button. This creates 200 realistic burn-in components across 5 lots with about 8% known defects.
- **Option 2 - CSV / Excel Upload tab:** click inside the dashed box, choose your own `.csv` or `.xlsx` file. The file should have at least: `component_id, lot_id, [parameter]_0h, [parameter]_24h, [parameter]_96h, [parameter]_168h`. The standard parameters are `iddq, leakage, delay` and `supply_current`.

After loading you see a data preview (columns and first rows) and a confirmation message with the record count.

2 **Run the analysis** — go to **Analysis Center** in the left menu.

You will see green text "*Dataset loaded: X components across Y lots - ready to analyze*". If data is missing, an orange warning appears and runs are blocked until you load data.

On the left there are four method tabs. Click **Comprehensive** first (it runs everything in one click):

1. Keep the default **Z-Score Threshold = 1.5** and **IQR Multiplier = 1.5** (recommended).
2. Click the **Run Comprehensive Analysis** button.
3. Wait a few seconds. The results appear below, including Module A verdicts, an **Anomaly Detection Score** and (after it finishes training) Module B drift predictions.

Alternative (optional) - run the two modules separately

In the same Analysis Center you can also use **Module A: Outlier** (lot-based outlier detection), **Module B: Drift** (Train Models + Run Batch Prediction), and **Single Prediction** (type a component's 0h and 24h values manually to predict its 168h value at once).

3 **Look at the results and certificates** — the results already appear on screen.

1. **Stat cards :** PASS / REVIEW / REJECT counts. Clicking a card *opens a certificate* underneath with the exact components.
2. **Detailed report :** in the results table, click **View Report** (or click a PASS/REVIEW/REJECT badge). A report panel slides in from the right showing raw test values at 0h/24h/96h/168h, per-parameter outlier checks, why-the-verdict reasons, and drift predictions to 168h.
3. **Certificate download :** inside the certificate, click **Download Certificate** to save an official-looking certificate file. Open it in a browser and print / save as PDF for a PDF copy.

4 **Check the Dashboard** — go to **Dashboard** in the left menu.

The mission-control screen summarises everything: total components, lots analyzed, anomalies found, drift models trained, the Module A PASS/REVIEW/REJECT split, the Anomaly Detection Score, and per-parameter drift accuracy (MAE / RMSE / R2) so you can judge how trustworthy the predictions are.

5. What Output You Will Get

Feature	Where to see it	What it shows
Data preview & confirmation	Upload Data page	File name, record count, column list, first rows of your file
Classification summary cards	Analysis Center	Total components and counts of PASS / REVIEW / REJECT
Anomaly Detection Score	Analysis Center & Dashboard	Score in % with TP / FP / TN / FN and Recall (sensitivity)
Component results table	Analysis Center (Comprehensive / Module A)	Each component: lot, classification badge, risk score, anomalies count, "View Report"

Certificate (Acceptance / Rejection / Manual Review)	After clicking a PASS / REVIEW / REJECT card	Certificate title, covered component list, test checklist, QA sign-off, reference number, Download button
Detailed component report	Click "View Report"	Raw 0h/24h/96h/168h values, per-parameter Module A analysis (z-score / IQR / absolute-limit flags), reasons & recommendations, Module B predicted 168h, safety-slope check
Drift accuracy metrics	Module B tab & Dashboard	Mean Absolute Error (MAE), RMSE, R2, MAPE per parameter, plus PASS/FAIL counts per prediction
Single prediction	Single Prediction tab	Predicted 168h value, drift rate, safety slope, exceed flag, recommendation, step-by-step explanation

6. What You Can Observe From the Output

6.1 Reading the classification cards

- **PASS (green)** - component is normal: all parameters are close to its lot average. Safe to use in the payload.
- **REVIEW (orange)** - a few parameters deviate moderately. Needs the QA inspector's manual check before a final decision.
- **REJECT (red)** - clearly abnormal values (outliers) detected. Must **not** be deployed.

6.2 Reading the Anomaly Detection Score

Formula: Score = max(0, (TP + TN - 10 x FN - 1 x FP) / Total)

- **TP** = defective parts that were correctly flagged. **TN** = good parts that were correctly passed.
- **FN** = defective parts that escaped (missed). A missing defective part is **catastrophic**, so one FN subtracts the value of 10 correct results.
- **FP** = good parts wrongly flagged; costs only 1 point (safe but wasteful).

Best scores are near 100%. On the included sample data the system typically scores around **81% with a 100% recall (zero escapes, FN = 0)** - very desirable in space applications.

6.3 Reading the drift (Module B) table

- Each row shows the **parameter**, its 0h and 24h inputs, the **Predicted 168h** value, the **Actual 168h** (when ground truth exists) and the **Prediction Error**.
- **Verdict PASS** means the predicted drift stays within the safety slope; **FAIL/REJECT** means the component drifts too fast for 168h - it should be removed early, before it fails in the field.
- A **low MAE** (for example delay MAE = 0.03) means the model is very accurate; small MAE relative to the measurement unit = trustworthy forecast.

6.4 Reading the detailed report

- **Burn-in dataset values** - the actual measured values across 0h / 24h / 96h / 168h.
- **Per-parameter checks** - for every parameter, three methods vote (Z-Score, IQR bounds, Absolute limit). FLAG / CLEAR tells you which method caught the problem.
- **Why this verdict** - plain-language reasons, e.g. "*Parameter 'leakage_0h' = 7.61 (lot mean 6.79, std 0.26) - flagged by 2/3 detection methods*", and recommendations such as "*Investigate oxide integrity - elevated leakage may indicate gate oxide degradation.*"
- **Module B drift to 168h** - predicted value, drift rate per hour, safety slope, and whether the component exceeds the safe drift.

Quick sanity checks to confirm results are sensible

1. PASS + REVIEW + REJECT should equal the total components you uploaded.
2. When you click a card, the certificate count should match the card number exactly.
3. On sample data, roughly 8% of parts are marked (REJECT + REVIEW) and the rest are PASS - matching the seeded defect rate.
4. FN should be 0 on the sample set (no defective part escapes).
5. Most MAE values should be small relative to the parameter's unit.

7. Common Problems & Solutions

Problem / Message	Solution
"No data loaded. Upload a CSV/Excel dataset..." (orange banner, buttons disabled)	Go to **Upload Data**, load the sample data or upload a file, then return to Analysis Center.
"Error: No parametric columns with time-series data found"	The file must contain `[parameter]_0h` and `[parameter]_168h` columns (example: `iddq_0h, iddq_168h`).
Login fails	Double-check the demo email/password from Section 3, or register a new account with a strong password.
Certificate looks wrong or download blocked	Download the certificate HTML, open it in a browser and use the browser's **Print > Save as PDF**.
Analysis empty / old results shown	Run **Run Comprehensive Analysis** again after loading data; results refresh each time.

Final tip - the 60-second demo

1. Log in with any demo account.
2. Upload Data tab: click **Generate & Load Sample Data**.
3. Analysis Center tab: click **Run Comprehensive Analysis**.
4. Click the **REJECT** card to open a Rejection Certificate, then **View Report** on any row to see the full explanation.
5. Open **Dashboard** to see the score and drift accuracy.